# Estimating Carbon – Tracking the Carbon Cost of AI in Development

*Max Davies, Iman Hadi, Morgan Parry, Hao Ni, Jacob Connor, Aayush Bhalerao*
*Client: UsTwo*

## Introduction

- ~90% of an AI model's emissions are produced during inference[1]
- AI used in IDEs has a hidden environmental cost that can be mitigated
- Development teams within the Green Software Foundation work towards an emissions-based budget using the Software Carbon Intensity Standard (SCI)[2]

## Our Goals

- To produce a prototype toolkit that plugs into development pipelines and estimates emissions in context
- To help teams make lower-carbon choices earlier on
- To contextualise AI emission costs in development

**It is now estimated that 80–90% of computing power** for AI is used for inference[1] - a hidden yet critical emission source in global climate governance

## Approach

- Analyse log files of built in AI assistants (Github Copilot) to retrieve metadata for each prompt.
- Intercept API calls made at runtime for any user-written files via proxy; analyse response metadata and pass response to user
- Apply emission calculation (see Methodology) to metadata
- Display latest emission with colour coded status bar
- Display log of emissions by call / prompt in side bar
- Present in-depth analysis within dashboard window

*From top to bottom: Timeline of carbon emissions, Session budget indicator, Average request graphic, Usage heatmap*

## Methodology

- Carbon emissions calculated in gCO2e; a standard measurement that converts different greenhouse gases into a metric representing their global warming potential relative to 1 gram of CO2
- Calculations align with the Green Software Foundation's methodology for Software Carbon Intensity (SCI)1 :

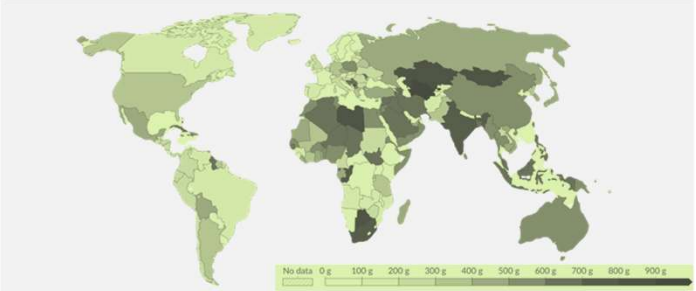
$$SCI = \left((E \times I) + M\right) \div R$$

Our methodology is below:

- Measure amount of processed input and output per prompt in tokens ($R$)
- Convert tokens into model-specific energy cost in kwh ($E$)
- Apply local grid carbon intensity factor to energy cost to produce carbon impact in gCO2e (I)
- Currently manufacturing emissions are not considered in our calculations ($M$)

## Accuracy of Estimation

- Since almost all AI companies refuse to publish data on their carbon or energy emissions during inference, our calculations are estimations using data collated from published research and articles.
- A major limitation faced with the SCI calculations was the lack of certainty of which AI data centre was processing our request. The Grid Carbon Intensity of a nation varies hugely; Sweden has a low 35 gCO2e / kwh whilst South Africa has a massive 713 gCO2e / kwh. As such, the global average of 471gCO2e/kwh was used.




*Lifecycle carbon intensity of electricity, 2025* [3]

University of BRISTOL

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**References**

1. *MIT Technology Review:* https://www.technologyreview.com/2025/05/20/1116327/ai-energy-usage-climate-footprint-big-tech
2. *Green Software Foundation:* https://sci.greensoftware.foundation/
3. *Ember Energy* https://ourworldindata.org/grapher/carbon-intensity-electricity

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